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Professional Summary

AI Automation Engineer shipping **reliable LLM-integrated workflows** with measurable outputs, human review steps, production monitoring, and evaluation hooks. Maps real business workflows to safe, documented automations — integrating Claude, Gemini 2.0 Flash, and MCP tool-calling with explicit fallbacks, validation layers, and model-boundary annotations. Delivered production systems to real clients: an offline-first retail POS (React + Electron + SQLite) and an enterprise multi-campus LMS (Next.js + PostgreSQL). Builds RAG retrieval pipelines, agent tool-calling interfaces, and FastAPI-backed automation stacks — and documents systems so clients and engineers can operate them confidently without ongoing developer dependency.

Core Competencies

Languages	Python, TypeScript, JavaScript, SQL, Bash
LLM Integration	Claude API, Gemini 2.0 Flash, OpenAI-compatible endpoints, LangChain, structured output design, prompt evaluation
Agent & Tool-Calling	MCP (Model Context Protocol), n8n workflow automation, agent instruction frameworks, tool-use orchestration, RAG pipeline design
APIs & Automation	FastAPI, Flask, REST API design, async endpoints, validation & fallback layers, structured data pipelines
Eval & Monitoring	Human-in-the-loop review, SHAP attribution, performance monitoring hooks, model-boundary safety annotations
Retrieval & Search	Chroma, sentence-transformers, vector embeddings, Tree-sitter AST parsing, semantic search
Databases & Data	PostgreSQL, MySQL, SQLite, SQLAlchemy, Prisma, Pandas, ETL pipeline design
Deployment & Ops	Docker, GitHub Actions, Vercel, Hugging Face Spaces, Electron (offline-first), MLflow

Shipped Systems — AI Automation & Client Delivery

CareVision — Multimodal Clinical AI Workflow

[Live](#) | [Source](#)

Gemini 2.0 Flash · FastAPI · PostgreSQL · SQLAlchemy · React 18 · TypeScript

- Integrated **Gemini 2.0 Flash** into a clinical PWA; designed **explainability-first human review workflows** with model-boundary annotations and confidence gates — preventing clinician overreliance by architecture, not by disclaimer.
- Built **validation and fallback layers** via SQLAlchemy-backed FastAPI: PostgreSQL persistence, offline-capable inference fallback, and step-level audit logging for full workflow traceability with zero silent failures.
- Applied **structured output schema enforcement** and prompt evaluation hooks throughout; submitted to the **Google Gemma 4 Hackathon**.

CodeScope MCP Preflight — Agent Tool-Calling & RAG Retrieval

[Source](#)

Python · MCP · Tree-sitter · sentence-transformers · Chroma · CLI

- Built a **local-first MCP server** serving repository intelligence as tool-callable context for AI agents: Tree-sitter AST parsing, sentence-transformers semantic embeddings, and Chroma vector retrieval — a complete **RAG pipeline** via the Model Context Protocol.
- Designed the tool-calling interface to enforce deliberate pre-generation decisions, reducing hallucinated or conflicting code paths in downstream agent outputs.

Mr. Stationer — Offline-First POS System

[Production Client Delivery](#)

React · Electron · Node.js · SQLite · Tiered Backup · Analytics Dashboard

- Designed, built, and deployed a complete **offline-first POS** for a real retail client: transaction processing, inventory management, tiered automatic backup, and an analytics dashboard — fully operational without internet connectivity.
- Owned the full client lifecycle — workflow scoping, architecture decisions, implementation, and **handoff documentation** so the client could operate the system independently without ongoing developer involvement.

AdaptIQ — Automated Business Demand Forecasting SaaS

[Live](#) | [Source](#)

Prophet · XGBoost · SARIMA · LSTM · FastAPI · Docker · Hugging Face Spaces

- Automated an end-to-end forecasting workflow over **580K+ retail transactions**; 4-model ensemble achieves **MAPE 1.23%** with **automated model selection** and automatic fallback to best-performing regime — measurable outputs with no manual override.
- Built **performance monitoring hooks**, confidence-interval scenario planning, and PDF/CSV report generation via FastAPI so non-technical clients consume automation outputs directly.

Evershaheen Academy — Enterprise Multi-Campus LMS

[Enterprise Client Delivery](#)

Next.js 15 · Prisma · PostgreSQL · Multi-Campus Architecture

- Built a multi-campus school management LMS for an institutional client; delivered **layered technical documentation** across

nine implementation layers so AI coding agents and client engineers could operate, extend, and maintain the system without developer dependency.

SentinelIQ — Industrial Predictive Maintenance Automation

[Live](#) | [Source](#)

PyTorch · TCN · LSTM · Isolation Forest · SHAP · PuLP · FastAPI · Docker

- Automated maintenance scheduling from sensor data: TCN-LSTM predicts remaining-useful-life (**RMSE 14.37**); PuLP scheduler converts predictions into optimal intervention windows — **\$38K attributed cost savings** with SHAP per-decision attribution for full auditability.
- Dual anomaly detection (Isolation Forest + Autoencoder) acts as a **validation and fallback layer**: conflicting signals trigger human review flags rather than auto-scheduling — production monitoring via FastAPI operator dashboard.

AI-Powered Lead Generation Workflow

[n8n Automation](#)

n8n · Google Gemini · Google Sheets · Gmail · Web Scraping

- Built an end-to-end automated lead pipeline in n8n: triggers on form submission, scrapes Google Maps, applies **duplicate filtering and quality classification**, and appends qualified prospects to Google Sheets — zero manual processing from trigger to storage.
- Integrated a **Gemini-powered AI Agent** with live web-scraping tools to enrich each lead: fetches website context, extracts contact emails, and generates personalised Gmail outreach; **explicit fallback routing** handles leads with no website or no email without silent failures.
- Closes with an automated **completion reporting step** that builds a structured campaign summary and dispatches a Gmail digest — full pipeline from lead capture through AI-enriched outreach with no manual intervention required.

AI-Powered WhatsApp Restaurant Chatbot

[n8n Automation](#)

n8n · Google Gemini Chat Model · WhatsApp Business API · Google Sheets · Simple Memory

- Built a production WhatsApp ordering chatbot in n8n: a **Gemini-powered AI Agent** with conversation memory (Simple Memory) handles inbound WhatsApp messages and routes each request to the correct tool — FAQ lookup, order placement, or inventory check.
- Designed a **three-tool agent interface** backed entirely by Google Sheets: FAQ (read), Post Orders (append to sheet), and Get Inventory (read) — customers browse menus, place orders, and check stock availability entirely within WhatsApp.
- Live execution log confirms **2.61s full pipeline** (agent inference: 1.13s, 655 tokens); bot correctly triggered on inbound WhatsApp message and returned a structured greeting with action menu in the verified execution output.

Additional Automation & Engineering Work

- AI Content Production Pipeline (@deepfx_ai)** — Built a Claude-powered content automation system with custom skill files; enforces slide schemas, accuracy gates, SEO metadata packages, and multi-platform distribution — repeatable output production from topic input alone.
- Agent Instruction Frameworks (SOUL.md / GEMINI.md)** — Authored domain-specific instruction frameworks for the Antigravity AI coding environment; enforces mandatory context loading, expert role assignment, security constraints, and token efficiency across agent sessions.
- VITAL-LINK — Multimodal LLM Workflow** — Chest X-ray, lung audio, and vitals fused via Gemini 2.0 Flash into a single assisted-review inference pipeline. [Source](#)
- COVID-19 Analytics Platform** — Automated WHO/JHU ETL into MySQL; FastAPI aggregation layer; fully Dockerised analytics stack with live monitoring. [Source](#)

Education

Bachelor of Science in Data Science

Sep 2022 – May 2026 (Expected)

GIFT University, Gujranwala, Pakistan

Relevant Coursework: Machine Learning, Deep Learning, Database Systems, Data Structures & Algorithms, Statistical Inference, Operations Research

Research: Co-author — *TopoLite-KD: Efficient Topology-Aware Knowledge Distillation for COVID-19 CT-Slice Classification* (manuscript in preparation)

Certifications & Continuous Learning

- Claude 101 Certification** | Anthropic *2026*
- AI Fluency Framework & Foundations** | Anthropic *2026*

Hackathons, Community & Leadership

- Google Gemma 4 Hackathon (2025)** — Submitted *CareVision*; multimodal clinical AI PWA with Gemini 2.0 Flash integration, human-review-first workflow design, and offline-capable inference architecture.
- AI/ML Content Creator** — Educational ML content on TikTok (@deepfx_ai) and Instagram (@explainable_ai); distils model architecture and applied AI concepts for technical and non-technical audiences — demonstrates clear written communication at scale.